

THE PAS, MB, Canada

WMO: 718670

Lat: 53.9747N

Lon: 101.0889W

Elev: 887

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-24.7	-37.7	0.7	-28.7	-33.4	0.9	-24.4	23.4	4.0	21.5	1.2	5.5	280	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	83.0	66.3	79.9	64.4	76.9	63.0	68.9	78.4	66.9	76.3	65.0	73.8	10.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	97.6	73.5	63.4	90.3	71.1	61.5	84.4	69.3	33.7	78.5	32.0	76.7	30.5	73.8	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	20.5	18.2	-35.4	89.7	4.7	3.1	-38.8	92.0	-41.5	93.8	-44.2	95.6	-47.6	97.9
			-35.6	73.4	4.7	2.9	-39.0	75.5	-41.7	77.2	-44.3	78.8	-47.7	80.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	-1.8	2.8	16.0	33.5	47.2	59.6	65.4	62.9	52.7	38.3	19.7	4.0		
	DBStd	25.96	14.57	13.36	14.15	11.43	9.05	7.06	5.49	6.04	7.83	8.06	11.79	14.44		
	HDD50	7423	1607	1322	1054	501	162	9	0	1	59	373	908	1427		
	HDD65	11646	2072	1742	1519	944	553	189	62	113	373	829	1358	1892		
	CDD50	1408	0	0	0	6	76	296	479	400	142	9	0	0		
	CDD65	156	0	0	0	0	2	26	76	47	5	0	0	0		
	CDH74	1338	0	0	0	0	46	255	599	382	56	0	0	0		
	CDH80	267	0	0	0	0	7	53	117	83	7	0	0	0		
	WSAvg	8.3	8.0	7.9	8.2	8.2	8.3	7.7	7.9	8.3	9.0	9.4	9.0	8.1		
	PrecAvg	18.5	0.8	0.6	0.8	1.0	1.5	2.7	3.0	2.5	2.3	1.5	1.0	0.8		
	PrecMax	23.9	2.4	2.5	2.6	3.8	3.9	7.6	8.3	5.8	5.1	4.7	2.4	2.0		
(15) Precipitation	PrecMin	12.6	0.2	0.0	0.0	0.0	0.3	0.8	0.5	0.3	0.2	0.0	0.2	0.0		
	PrecStd	3.0	0.5	0.4	0.5	0.7	0.9	1.5	1.6	1.4	1.2	1.0	0.6	0.4		
	0.4%	DB	37.5	38.7	49.2	68.0	79.2	85.6	87.1	86.9	80.1	65.7	47.9	36.8		
	MCWB	34.6	33.9	40.1	51.0	57.4	66.5	69.3	68.6	63.7	53.2	40.6	33.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	30.4	33.5	44.0	61.3	73.2	80.4	83.2	81.9	74.5	59.0	42.0	30.9		
	MCWB	28.4	31.0	37.0	47.8	55.5	63.4	66.8	66.0	61.3	50.0	37.4	28.6			
	5%	DB	24.8	28.3	39.8	55.1	68.7	76.5	80.2	78.3	69.9	54.3	37.6	26.7		
	MCWB	23.3	26.2	34.8	44.2	53.0	61.3	65.1	64.8	58.6	47.5	34.7	25.2			
	10%	DB	20.3	23.3	35.6	50.1	63.8	72.8	77.3	74.6	65.4	50.1	34.5	23.4		
	MCWB	19.2	21.6	32.0	41.0	50.9	59.4	63.7	62.9	56.9	45.1	32.3	22.2			
	0.4%	WB	34.6	34.6	40.8	52.3	61.2	69.8	72.4	71.2	66.5	55.3	41.7	33.4		
	MCDB	37.2	37.8	48.1	65.2	75.0	77.6	81.7	82.2	76.7	63.2	46.0	36.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.4	31.0	37.5	48.5	57.2	66.1	69.4	68.4	62.7	51.1	37.7	28.7		
	MCDB	30.3	33.4	43.1	60.2	68.5	76.6	78.6	78.4	71.8	57.5	41.1	30.7			
	5%	WB	23.4	26.3	35.0	44.9	54.5	63.6	67.3	66.4	60.1	48.0	34.8	25.2		
	MCDB	24.8	28.3	39.4	54.4	65.6	72.9	76.5	75.4	68.2	53.4	37.5	26.5			
	10%	WB	19.3	21.6	32.3	41.4	51.9	61.2	65.6	64.3	57.3	45.2	32.3	22.3		
	MCDB	20.3	23.2	35.5	49.1	62.5	69.8	74.5	72.2	64.3	49.5	34.4	23.5			
	MDBR	15.1	18.3	20.1	19.8	20.6	18.6	18.3	17.6	16.7	12.6	11.7	13.5			
	5% DB	MCDBR	17.8	18.5	20.5	26.3	28.8	24.1	23.1	23.4	24.1	20.8	14.3	15.4		
(36) Mean Daily Temperature Range	5% WB	MCWBR	16.4	15.9	14.7	15.9	14.7	10.9	10.5	11.3	13.2	12.8	10.9	13.9		
	MCDBR	17.6	17.7	19.5	25.5	25.4	20.6	20.3	20.5	22.1	19.0	13.5	15.0			
	MCWBR	16.3	15.3	14.4	15.7	14.0	10.6	10.4	11.0	12.9	12.3	10.5	13.7			
	taub	0.235	0.244	0.272	0.337	0.348	0.378	0.389	0.363	0.321	0.289	0.246	0.221			
(40) Clear-Sky Solar Irradiance	taud	2.306	2.334	2.360	2.372	2.424	2.361	2.380	2.448	2.553	2.576	2.441	2.313			
	Ebn at Noon	241	279	292	283	285	276	271	272	273	260	239	225			
	Edh at Noon	17	24	29	34	34	36	35	31	25	19	15	14			
	RadAvg	299	659	1138	1572	1744	1770	1802	1478	1024	564	263	196			
(44) All-Sky Solar Radiation	RadStd	29	43	93	122	170	134	114	114	112	76	33	26			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+2.68	N/A	-0.68	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	+2.92	N/A	-0.65	-0.74	N/A	N/A	N/A	N/A

Nomenclature: See separate page